A Worked example at $E = E^{\circ\prime}$

Given: $E = E_{0'} = -250$ mV, $k^0 = 127$ s^{-1}

$\alpha = 0.37$, $n_e = 2$, $f = 15.0$ Hz, $\tau = 33.3$ ms

Step 1: Forward pulse: $E_+ = E_0 + E_{sw} = -225$ mV

$\xi_+ = n_e F(E_+ - E_0)/RT = 1.946$

$k_f = k^0 e^{-\alpha\xi_+} = 61.9$ s^{-1}

$k_b = k^0 e^{(1-\alpha)\xi_+} = 433.6$ s^{-1}

Step 2: $\Gamma_{eq}^{(+)} = k_f/(k_f + k_b) = 0.1250\,\Gamma_{tot}$

$\Gamma_{red}(\tau) = \Gamma_{eq} + (\Gamma_{red}(0) - \Gamma_{eq})e^{-(k_f+k_b)\tau}$

$= 0.1250 + (0.5 - 0.1250)\cdot 0.0000$

$= 0.1250\,\Gamma_{tot}$

Step 3: Backward pulse: $E_- = -275$ mV; IC = Step 2 result

$\Gamma_{red}(\tau) = 0.8750\,\Gamma_{tot}$

Step 4: $I_f = 634.6387$ nA, $I_b = 2678.7573$ nA

$\Delta I = I_f - I_b = -2044.1186$ nA





